

Name:

Date:

Period:

Newton #4 (Weight & Vertical Motion)

$F_{NET} = ma$ $F_{NET} = \pm F_1 \pm F_2 \pm \dots$ $F_G = mg$ $g = 9.8 \text{ m/s}^2$

1. *Horizontal Review Problem:* A 1250-kg car is propelled by an applied force 4500 N. The force of friction is 212 N and there is 490 N of drag. Calculate the car's acceleration.

2. Consider a 12-kg bowling ball.
 - (a) What is the bowling ball's weight on earth?

 - (b) What is the bowling ball's weight on Mars where $g = 3.71 \text{ m/s}^2$?

3. What is the normal force acting on a 65-kg man standing in an elevator that is accelerating downward at a rate of 3.1 m/s^2 ?

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4. Bobby is lifting buckets of water straight up out of a well using a rope. If he applies a force of 50 N to a bucket with a mass of 3.2 kg, what is the acceleration of the bucket?

5. In the famous Leaning Tower of Pisa experiment, Galileo dropped two balls from the top of the tower. Let's say that one was 5 kg and the other was 500 kg

(a) Calculate the weight of the 5-kg ball.

(b) Calculate the weight of the 500-kg ball.

- (c) Newton's Second Law says that the ball with more force (*i.e.* more weight) should have a greater acceleration. How can both balls have the same acceleration?

